



Verified Carbon Standard

TAIBA N'DIAYE WIND POWER



Document Prepared By: 4K Earth Science Private Limited

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Summary:

AERA Group has commissioned “4K Earth Science Private Limited” to carry out the Joint Validation and Verification of the project “Taiba N’Diaye Wind Power”, with regard to the relevant requirements of VCS Standard Version 4.1. The monitoring period verified is from 09/12/2019 to 31/07/2021 (including both the days).

The project activity involves the installation and operation of a 158.7 MW wind farm in the commune of Taiba N’Diaye, Senegal. The project activity which is Senegal’s first utility-scale wind farm, consists of 46 Vestas V126 wind turbines, for a total installed capacity of 158.7 MW. They utilise a 117m tubular steel tower and a blade length of 61.7m giving a very large swept area of 12,469 m², allowing the wind turbines to maximise the amount of energy captured from the wind, and are designed for at least 20 years of operation.

The main elements of the wind power plant are Vestas V126-3.45 MW turbines, pitch regulated upwind turbines with active yaw and a three blade rotor. The wind turbine family utilises the OptiTip® concept and a power system based on an induction generator and full-scale converter. With these features, the wind turbine is able to operate the rotor at variable speed and thereby maintain the power output at or near rated power even in high wind speed. The technical details of the project is verified from the completion certificate /17/, technical specification of wind turbines /18/, single line diagram /20/, signed PPA /26/, photos /25/ and during remote audit and found consistent.

The project has been built by Lekela and is owned and operated by Parc Eolien Taiba N’Diaye S.A.U.

PETN is designed to generate electricity for at least 20 years through its 46 Vestas wind turbines. The generated electricity is supplied to SENELEC, the state-owned utility, under a long-term power purchase agreement (PPA) /26/.

In the absence of the project activity, electricity would have been generated by the operation of grid connected power plants, which are predominantly fossil fuel based. Therefore Taiba N’Diaye Wind Power project #2588 displaces carbon dioxide that would have been produced in the absence of the project activity by the grid connected power plants.

It has been estimated that approximately 450 GWh of electricity will be generated annually by the project (equivalent to 2,834 hours/year or 32% load factor), displacing approximately 243,695 tonnes of carbon dioxide (tCO₂) per annum and over 2.4 million tCO₂ over the 10-years crediting period.

Project is fully commissioned as verified from the completion certificates /17/.

Validation and Verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified emission reductions (VERs).

A risk-based approach has been followed to perform this joint validation and verification. In the course of the joint validation and verification, 03 Corrective Action Requests (CAR) and 04 Clarification Requests (CL) were raised and successfully closed out. 00 Forward action request (FARs) was raised.

The joint validation and verification is based on the Joint PD & MR, additional documents related to baseline and monitoring methodology, ER calculation sheet, the subsequent background investigation, follow-up interviews and supporting documents made available to the validation and verification team by project proponent.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. As a result of the validation, the validation team confirms that:

- The project fulfils criteria of VCS Standard Version 4.1.
- The project is in line with all relevant VCS requirements.
- The project additionality is sufficiently justified in the Joint PD & MR.
- The monitoring plan is transparent, adequate and in line with applied baseline and monitoring methodology ACM0002 version 20.0.
- The calculation of the Baseline emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 2,436,950 tCO₂e is most likely to be achieved within the 10 years of crediting period.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria.

No restrictions or uncertainties were identified related to the joint validation and verification.

The management of the 'Parc Eolien Taiba N'Diaye S.A.U' responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions on the basis set out within the project final JOINT PROJECT DESCRIPTION & MONITORING REPORT Version 2.1 dated 14/01/2022. The calculation and determination of GHG emission reductions from the project is the responsibility of the management of the Aera Group. The development and maintenance of records and reporting procedures are in accordance with the JOINT PROJECT DESCRIPTION & MONITORING REPORT Version 2.1 dated 14/01/2022.

It is our responsibility to express an independent GHG verification opinion on the GHG emissions and on the calculation of GHG emission reductions from the project for the period 09/12/2019 to 31/07/2021 based on the reported emission reductions in the final JOINT PROJECT DESCRIPTION & MONITORING REPORT Version 2.1 dated 14/01/2022 for the same period.

Further during this monitoring period (09-December-2019 to 31-July-2021), the project has produced a net total of 537,416 MWh electricity and a total amount of 307,939 tCO₂ of emission reduction. The monitoring period subject to this monitoring report is inclusive of first and last day of period.

Based on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these, 4KES planned and performed our work to obtain the information and explanations that we considered necessary to provide sufficient evidence for us to give reasonable assurance that this reported amount of GHG emission reductions for the period is fairly stated.

4KES confirms the following;

Reporting period: 09/12/2019 to 31/07/2021

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2019 (from 09/12/2019)	13,142	0	0	13,142
2020	124,066	0	0	124,066
2021 (until 31/07/2021)	170,731	0	0	170,731
Total	307,939	0	0	307,939

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1 INTRODUCTION

1.1 Objective

4KES has been contracted by *AERA Group* to undertake the Joint validation and verification of the project titled “Taiba N'Diaye Wind Power” (VCS ID 2588). The monitoring period verified is from 09/12/2019 to 31/07/2021 (including both the days).

The purpose of this joint validation and verification is to have an independent third party assessment of whether the project activity conforms to the qualification criteria set out in the VCS standard Version 4.1 /08/ to attain real, measurable, additional and permanent emission reductions.

The validation statement/opinion is a written assurance that:

- The project complies with all the applicable VCS requirements and has the ability to generate the emission reductions stated over the project’s crediting period.
- The validation followed the requirements of the current version of the VCS Standard Version 4.1 /08/ and VCS program guide 4.0 /07/ to ensure the quality and consistency of the validation work and the report.

The objectives of the verification exercise are, by review of objective evidence, to establish that:

- The project activity has been implemented and operated as per the project description (PD) and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- Monitoring report and other supporting documents are complete;
- The data is recorded and stored as per the monitoring methodology and approved monitoring plan.
- To confirm that the monitoring system is implemented and fully functional to generate Voluntary Emission Reductions (VERs/VCUs) without any double counting, and
- To establish that the data reported are accurate, complete, consistent, transparent and free of material error or omission by checking the monitoring records and the emissions reduction calculation.

1.2 Scope and Criteria

The validation scope is given as an independent and objective review of the project design and monitoring plan which is included in the Joint PD & MR and other relevant supporting documents.

The scope of work covered in the validation is described below:

- To validate whether the project activity meets the requirements of VCS Standard Version 4.1 /08/ and VCS program guide 4.0/07/ including additionality and compliance with local laws.
- To evaluate whether the baseline and monitoring plan are in conformance with the applied methodology from the VCS approved GHG program.
- To confirm that the information presented are completed, consistent, transparent and free of omission or material error.

- Background investigation and follow up interviews.

The information in the Joint PD & MR is reviewed against the criteria of VCS Standard 4.1 /08/, VCS program guide 4.0 /07/ and the approved baseline and monitoring CDM methodology ACM0002 version 20.0.

4KES has performed validation based on a risk based approach focusing mainly on the significant risks to meet the qualification criteria and the ability to generate Verified Carbon Units (VCUs).

The scope of verification is to assess the claims and assumptions made in the Joint PD & MR against the VCS criteria, including but not limited to, VCS standard, applied methodology and other relevant rules and requirements established for VCS project activities.

The validation *and verification* is not meant to provide any consulting towards the client. However, stated request for clarifications and/or corrective actions may provide input for improvement of the project design.

1.3 Level of Assurance

The Joint validation & verification report is based on *Joint PD & MR /02/*, Monitoring plan, supporting documents made available to the validator and information collected through performing interviews during the remote audit. Based on the process and procedures conducted, 4KES states with a reasonable level of assurance that information in the *Joint PD & MR*:

- is materially correct and is a fair representation of the actual project details, and
- is prepared in accordance with VCS requirements and the applied CDM methodology for information pertaining to additionality, GHG qualification, monitoring and reporting

The validation & verification work is carried out as per this requirement and the validation & verification opinion is assured provided the credibility of all above. Details are presented in the Validation & Verification statement in section 4 below.

The verification team verified the complete monitoring data /21/, /22/, /23/ and /25/ for the monitoring parameter of the monitoring plan against the *Joint PD & MR /02/* and ER Calculation sheet /04/ and confirms that the reported emission reductions are free from any type of material errors. Remote audit was also conducted to verify the implementation and monitoring plan of the project activity. The implementation of the project activity (i.e. major equipment and the metering arrangement) was verified during the Microsoft Teams remote interview and found in accordance with the *Joint PD & MR /02/*. Verification team has also verified the technical details of the project equipments and metering arrangement with the supportive documents /18/, /20/ and /25/ and found correct. Therefore, 4KES confirms that the validation & verification is conducted with reasonable level of assurance.

1.4 Summary Description of the Project

The project activity involves the installation and operation of a 158.7 MW wind farm in the commune of Taiba N'Diaye, Senegal. The project activity which is Senegal's first utility-scale wind farm, consists of 46 Vestas V126 wind turbines, for a total installed capacity of 158.7 MW. They utilise a 117m tubular steel tower and a blade length of 61.7m giving a very large swept area of 12,469 m², allowing the wind turbines

to maximise the amount of energy captured from the wind, and are designed for at least 20 years of operation.

The main elements of the wind power plant are Vestas V126-3.45 MW turbines, pitch regulated upwind turbines with active yaw and a three blade rotor. The wind turbine family utilises the OptiTip® concept and a power system based on an induction generator and full-scale converter. With these features, the wind turbine is able to operate the rotor at variable speed and thereby maintain the power output at or near rated power even in high wind speed. The technical details of the project is verified from the completion certificate /17/, technical specification of wind turbines /18/, single line diagram /20/, signed PPA /26/, photos /25/ and during remote audit and found consistent.

Project is Senegal's first large scale wind energy project and a critical part of the government's "Plan Senegal Emergent". The project has been built by Lekela and is owned and operated by Parc Eolien Taiba N'Diaye S.A.U.

PETN is designed to generate electricity for at least 20 years through its 46 Vestas wind turbines. The generated electricity is supplied to SENELEC, the state-owned utility, under a long-term power purchase agreement (PPA) /26/.

In the absence of the project activity, electricity would have been generated by the operation of grid connected power plants, which are predominantly fossil fuel based. Therefore Taiba N'Diaye Wind Power project #2588 displaces carbon dioxide that would have been produced in the absence of the project activity by the grid connected power plants.

The project implementation took place in three phases:

1. the first phase with the installation of 16 wind turbines i.e. a total power of 55.2 MW,
2. the second phase with the installation of 16 wind turbines with a total power of 55.2 MW,
3. the third tranche with the installation of 14 wind turbines with a total power of 48.3 MW.

It has been estimated that approximately 450 GWh of electricity will be generated annually by the project (equivalent to 2,834 hours/year or 32% load factor), displacing approximately 243,695 tonnes of carbon dioxide (tCO₂) per annum and over 2.4 million tCO₂ over the 10-years crediting period.

The implementation timeline for the project is as follows as verified from supportive documents /28/, /30/, /26/ and /17/. Project is fully commissioned as verified from the completion certificates /17/.

22-28 December 2012	Public Stakeholders Meeting
21 June 2016	Environmental Impact Clearance
10 October 2016	PPA signature
09 December 2019	Phase 1 Commissioning date
06 November 2020	Phase 2 Commissioning date
30 June 2021	Phase 3 Commissioning date

The project participants involved in the project are as per below tables.

Project participant:

Organization name	Parc Eolien Taiba N'Diaye S.A.U
Contact person	M. Alioune BA
Title	Finance Director
Address	Immeuble Riviona 167 Avenue Lamine Gueye Dakar
Telephone	+221 33 849 73 93
Email	taiba@lekela.com

Other Entities Involved in the Project:

Organization name	SENELEC
Role in the project	Public utility & conceding authority
Contact person	M. Papa Toby GAYE
Title	Head of properties
Address	28 rue Vincens BP93 Dakar Senegal
Telephone	+221 33 839 30 30
Email	toby.gaye@senelec.sn

Organization name	Aera Group
Role in the project	Carbon consultant & offtaker
Contact person	M. Alexandre DUNOD

Title	Head of Certification and Deliveries
Address	28 cours Albert 1er 75008 Paris France
Telephone	+33 142 180 202
Email	a.dunod@aera-group.fr

Parc Eolien Taiba N'Diaye S.A.U has gained project ownership through a production license /19/and power purchase agreement /26/ with Senegalese public utility SENELEC.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The validation and verification process involves the following:

- The desk review of documents and evidences submitted by the project proponent in context of the reference VCS rules and guidelines,
- Undertaking remote audit, interview or interactions with the representative of the project proponent,
- Reporting audit findings with respect to clarifications and non-conformities and the closure of the findings, as appropriate and
- Preparing a draft validation and verification report
- Resolution of outstanding issues and the issuance of final validation and verification report and opinion
- The prepared validation & verification report and other supporting documents then undergo an internal quality control before being submitted to the Verra Secretariat for registration & issuance of credits as per VCS standard version 4

In order to ensure transparency, a validation & verification protocol was prepared for the project according to the VVS for PAs version 03.0 *validation & verification requirements* and VCS Standard version 4.1 /08/. The Clarification Requests (CL) were issued where additional information was needed to clarify issues, and Forward Action Requests (FAR) for issues relating to project implementation that required review during the *next* verification of the project activity.

4KES assessed and determined whether the proposed implementation and operation of the project activity, and the steps taken to report emission reductions comply with the criteria and relevant guidance provided by the VCS Board.

Duration of Validation:

Validation & Verification Contract	27/08/2021
Remote audit	26/10/2021
Findings raised	29/10/2021
Draft Validation and Verification Report	20/11/2021
Final Validation and Verification Report	15/01/2022

2.2 Document Review

The validation & verification has been conducted using the VCS Standard and the applied methodology ACM0002 version 20.0 as the reference criteria. The validation team had done the completeness check of Joint PD & MR submitted by the PP as per the VCS standard Version 4.1 /08/ requirements. Furthermore, a desk review was also carried out to assess the following:

- Information of project details in compliance with VCS PD template
- Appropriateness of methodology ACM0002 version 20.0 applied to the project activity
- Compliance with relevant laws and regulations
- Correctness of application of baseline and monitoring methodology
- Demonstration of additionality
- Monitoring plan described in the VCS PD
- Stakeholder consultation
- Proof of listing of project under pipeline
- Calculation of grid emission factor, etc. where applicable.

The Joint PD & MR version 1.0 dated 26/08/2021 /01/ was initially reviewed and the PP requested to submit the revised documents along with the supporting information and documents. The revised documents and additional supporting documents were further assessed by the validation & verification team during desk review. During the validation and verification process, the revised Joint PD & MR (version 2.1, dated 14/01/2022) /02/ and the supporting documents were assessed to confirm the actions taken by the PP to the CARs and CLs issued.

The validation team has reviewed the final version of the Joint PD & MR (i.e. Version 2.1 dated 14/01/2022) to confirm that all changes agreed have been incorporated adequately.

Further, prior to the remote audit, it was verified by the validation team that the project was listed in the VCS pipeline in line with the requirements (Section 3, Registration and Issuance process, version 4.0 /09/) Ref: <https://registry.verra.org/app/projectDetail/VCS/2588>

The documents that were considered during the validation *and verification* process are given in Appendix of this report.

The assessment is performed by a verification team using a protocol. The cross checks between information provided in the Joint PD & MR and information from sources other than those used, if available, the team's sectoral or local expertise and, if necessary, independent background investigations.

2.3 Interviews

Due to worldwide COVID-19 spread and air travel restrictions, Validation & Verification team could not conduct the site visit. Since the date of closure of lockdown is uncertain due to the increase spread of COVID-19 and hence VVB did not conduct site visit for this project activity. However, the team performed the Microsoft Teams/Email remote interview with the PP representative, site person, consultant and

stakeholder and reviewed documents to achieve a reasonable level of assurances. This is in line with Section 4.1.2 of the VCS Standard, v4.1 which does not explicitly mandate site visits as part of the validation and verification process, only that VVBs must achieve a reasonable level of assurance on all validations and verifications. No sampling procedures were adopted. Kindly find below names of the persons interviewed (Microsoft Teams/Email remote interview).

Location	Country:	Senegal
	Region:	Thies
	Department:	Tivaouane
	Municipality	Taïba Ndiaye
	Locations under the influence of the project	Ndomor, Keur Malé, Minam, Mbayéne, Keur Birama, Keur Samba Awa, Keur Mbaye Sénoba. Taïba Mbaye, Same Ndiaye, Baïty Ndiaye, Baïty Gueye, Keur, Madiagne, Taïba Santhie, Keur Assane
Dates	26/10/2021 (Microsoft Teams interview/Email interview)	
Key points discussed	Name of person, interviewed	Designation, Organization
Implementation, Monitoring, Operational data, Calibration, Data collection, QA/QC procedures, Calculation of ERs, VCS requirements, Stakeholder meeting, on-going mechanism, Sustainable development Impact, Complaints, Environmental impact	Moustapha Thiam	GM, Parc Eolien Taiba N'Diaye S.A.U
	Alioune Ba	Finance Director, Parc Eolien Taiba N'Diaye S.A.U
	Yassine Majdallah	Plant Manager, Parc Eolien Taiba N'Diaye S.A.U
	Amadou Sow	ESG Manager, Parc Eolien Taiba N'Diaye S.A.U
	Alexandre Dunod	Chief Operating Officer, Director of Operations, AERA group
	Monsieur Aly Lo	Taiba Ndiaye village mayor's

2.4 Site Inspections

Due to worldwide COVID-19 spread and air travel restrictions, Validation & Verification team could not conduct the site visit. Since the date of closure of lockdown is uncertain due to the increase spread of COVID-19 and hence VVB did not conduct site visit for this project activity. However, the team performed the Microsoft Teams/Email remote interview with the PP representative, site person, consultant, stakeholder and reviewed documents to achieve a reasonable level of assurances. This is in line with Section 4.1.2 of the VCS Standard, v4.1 which does not explicitly mandate site visits as part of the validation and verification process, only that VVBs must achieve a reasonable level of assurance on all validations and verifications.

2.5 Resolution of Findings

As an outcome of the validation and verification process, the team can raise different types of findings:

A Clarification Request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met

Where a non-conformance arises the team leader shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions;
- The VCS requirements have not been met;
- There is a risk that emission reductions cannot be monitored or calculated.
- The validation and verification process may be halted until this information has been made available to the team leader's satisfaction. Failure to address a CL may result in a CAR. Information or clarifications provided as a result of a CL may also lead to a CAR.

During the validation and verification process, total 03 CAR and 04 CL were raised and resolved satisfactorily. The list of CARs/CLs/FARs raised and the response provided, the mean of validation and verification, reasons for their closure and references to correction in the relevant documents are provided in Appendix of this report.

2.5.1 Forward Action Requests

No FAR raised during the validation & verification.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity involves the installation and operation of a 158.7 MW wind farm in the commune of Taiba N'Diaye, Senegal. The project activity which is Senegal's first utility-scale wind farm, consists of 46 Vestas V126 wind turbines, for a total installed capacity of 158.7 MW. They utilise a 117m tubular steel tower and a blade length of 61.7m giving a very large swept area of 12,469 m², allowing the wind turbines to maximise the amount of energy captured from the wind, and are designed for at least 20 years of operation.

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Project is Senegal's first large scale wind energy project and a critical part of the government's "Plan Senegal Emergent". The project has been built by Lekela and is owned and operated by Parc Eolien Taiba N'Diaye S.A.U.

PETN is designed to generate electricity for at least 20 years through its 46 Vestas wind turbines. The generated electricity is supplied to SENELEC, the state-owned utility, under a long-term power purchase agreement (PPA) /26/.

In the absence of the project activity, electricity would have been generated by the operation of grid connected power plants, which are predominantly fossil fuel based. Therefore Taiba N'Diaye Wind Power project #2588 displaces carbon dioxide that would have been produced in the absence of the project activity by the grid connected power plants.

The project implementation took place in three phases:

4. the first phase with the installation of 16 wind turbines i.e. a total power of 55.2 MW,
5. the second phase with the installation of 16 wind turbines with a total power of 55.2 MW,
6. the third tranche with the installation of 14 wind turbines with a total power of 48.3 MW.

It has been estimated that approximately 450 GWh of electricity will be generated annually by the project (equivalent to 2,834 hours/year or 32% load factor), displacing approximately 243,695 tonnes of carbon dioxide (tCO₂) per annum and over 2.4 million tCO₂ over the 10-years crediting period.

The implementation timeline for the project is as follows as verified from supportive documents /28/, /30/, /26/ and /17/. Project is fully commissioned as verified from the completion certificates /17/.

22-28 December 2012	Public Stakeholders Meeting
21 June 2016	Environmental Impact Clearance
10 October 2016	PPA signature
09 December 2019	Phase 1 Commissioning date

06 November 2020	Phase 2 Commissioning date
30 June 2021	Phase 3 Commissioning date

Operation and maintenance of the project activity is taken care by VESTAS as checked from the contract /33/.

Project comes under Sectoral scopes: Energy industries (renewable - / non renewable sources) (Sectoral Scope 1). VVB has this sectoral scope to validate this project.

As the project is a Grid-connected electricity generation using wind and located in Senegal which is a Least Developed Country (as designated by United nations <https://unfccc.int/topics/resilience/workstreams/national-adaptation-programmes-of-action/ldc-country-information>), it is eligible under the VCS in accordance with table 1 of the VCS Standard, v4.1 /08/.

Project is not a group project activity as confirmed from section 1.4 of the VCS PD /02/ and during the remote interview.

Validation team has done the remote audit on 26/10/2021 to confirm the project description mentioned in the VCS-PD/02/.

The project participants involved in the project are as per below tables.

Project participant:

Organization name	Parc Eolien Taiba N'Diaye S.A.U
Contact person	M. Alioune BA
Title	Finance Director
Address	Immeuble Riviona 167 Avenue Lamine Gueye Dakar
Telephone	+221 33 849 73 93
Email	taiba@lekela.com

Other Entities Involved in the Project:

Organization name	SENELEC
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Role in the project	Public utility & conceding authority
Contact person	M. Papa Toby GAYE
Title	Head of properties
Address	28 rue Vincens BP93 Dakar Senegal
Telephone	+221 33 839 30 30
Email	toby.gaye@senelec.sn

Organization name	Aera Group
Role in the project	Carbon consultant & offtaker
Contact person	M. Alexandre DUNOD
Title	Head of Certification and Deliveries
Address	28 cours Albert 1er 75008 Paris France
Telephone	+33 142 180 202
Email	a.dunod@aera-group.fr

Parc Eolien Taiba N'Diaye S.A.U has gained project ownership through a production license /19/and power purchase agreement /26/ with Senegalese public utility SENELEC.

The start date of the project is 09-December-2019 (as the date of commissioning of PETN first tranche). Validation team has checked the completion certificate /17/ to confirm and found the start date in compliance with section 3.7 of the VCS standard Version 4.1 /08/. Also as per section 3.7.1 of the VCS standard version 4.1 /08/, projects shall complete validation within two years of the project start date which is also fulfilled.

The first crediting period of project activity starts from 09-December-2019 until 08-December-2029 (10 years fixed) in accordance with section 3.8.1 of the VCS Standard Version 4.1 /08/.

Project come under the scale “Project” as the estimated GHG emissions reductions per years are less than 300,000 tonnes of CO₂e per year in accordance with section 3.9.1 of the VCS Standard Version 4.1 /08/.

It has been estimated that approximately 450 GWh of electricity will be generated annually by the project (equivalent to 2,834 hours/year or 32% load factor), displacing approximately 243,695 tonnes of carbon dioxide (tCO₂) per annum and over 2.4 million tCO₂ over the 10-years crediting period.

Project Location is as follows which is verified from the web search.

Country:	Senegal
Region:	Thies
Department:	Tivaouane
Municipality	Taïba Ndiaye
Locations under the influence of the project	Ndomor, Keur Malé, Minam, Mbayéne, Keur Birama, Keur Samba Awa, Keur Mbaye Sénoba. Taïba Mbaye, Same Ndiaye, Baïty Ndiaye, Baïty Gueye, Keur, Madiagne, Taïba Santhie, Keur Assane The project site's geo-coordinates are: Latitude 15° 00'36.5"N, Longitude 16° 51'32.2"W

The project activity is in compliance with the Local Laws and regulations as confirmed from the document /30/ and /31/.

The project neither has nor intended to generate any other form of GHG related environmental credit for the GHG emission reduction or removal claimed under the VCS programme. Although it was originally registered under the CDM by its former developers¹, it was postponed and re-designed by its new developers, for certification and verification under the VCS only.

Validation and Verification team has checked the project registration under CDM (<https://cdm.unfccc.int/Projects/DB/DNV-CUK1330446091.13/view>) which is not able to renew anymore and no verification performed. Declaration /15/ for double counting has been checked and found OK. Further project is submitted for deregistration from CDM as checked from submitted form /16/, hence there is no double counting associated with project activity being participation of other GHG programs.

¹ As per CDM Project 5846: Taiba N'diaye Wind Energy project, Senegal registered on 29 Feb 12, undergoing de-registration

No commercially sensitive information has been excluded from the public version of the project description as verified during the remote audit.

Validation team confirms that the description in the project description is accurate, complete, and provides an understanding of the nature of the project and the project has been implemented as described in the project description.

Sustainable development contributions from the project activity has been mentioned under section 1.17 of the VCS PD /02/ and found OK. It is further verified from the supportive document /29/ and remote interview with stakeholder. Project contribute positively in the sustainable development of the area.

3.2 Participation under Other GHG Programs

The project neither has nor intended to generate any other form of GHG related environmental credit for the GHG emission reduction or removal claimed under the VCS programme. Although it was originally registered under the CDM by its former developers², it was postponed and re-designed by its new developers, for certification and verification under the VCS only.

Validation and Verification team has checked the project registration under CDM (<https://cdm.unfccc.int/Projects/DB/DNV-CUK1330446091.13/view>) which is not able to renew anymore and no verification performed. Declaration /15/ for double counting has been checked and found OK. Further project is submitted for deregistration from CDM as checked from submitted form /16/, hence there is no double counting associated with project activity being participation of other GHG programs.

3.3 Safeguards

3.3.1 No Net Harm

Being a renewable project activity, No negative impacts from the project activity. Wind energy systems do not produce air pollution or greenhouse gases. In general, using wind energy has a positive, indirect effect on the environment when wind energy replaces or reduces the use of other energy sources that have larger effects on the environment, i.e., there is not net harm.

PP has clarified the all the potential environment and socio-economic impacts and mitigation measure for them in detail under section 2.1 of the VCS PD /02/ which is found OK. The Project activity meets all local laws and regulation of Senegal.

Further socio-economic impacts are rather indirect but positively related to the project activity. There is no negative socio-economic impacts from the project which is also discussed during the stakeholder interview.

² As per CDM Project 5846: Taiba N'diaye Wind Energy project, Senegal registered on 29 Feb 12, undergoing de-registration

Project has been granted an Environmental Impact Assessment approval by decree on 21-June-2016 as verified from the supportive document /30/. Validation team has interviewed the local stakeholder and confirmed that project has positive impact on the environment and proper compensation was given to people.

3.3.2 Local Stakeholder Consultation

Local Stakeholders Consultation took place from 22 to 28 December 2012, between Dakar and Thiès, where the following actors were consulted:

- *The National Technical Departments:*
 - the Directorate of Environment and Classified Establishments;
 - the Civil Protection Directorate;
- *the technical services of the Thiès region:*
 - the Regional Division of the Environment and Classified Establishments;
 - the Regional Directorate of Rural Development;
 - the Regional Water and Forestry Inspectorate;
 - Regional Development Agency;
 - the Regional Planning Department;
- *the Sub-Prefect of Ouadiour;*
- *the Commune of Taïba N'diaye;*
- *the populations of the villages of Diambalo, Balsande, Taïba N'diaye, Taïba Mbaye, Baïty N'diaye, Baïty Guèye, Minam Diop, Mbayène, Khelkom Diop, Taïba Santhie, Maka Gaye Bèye and Ndomor Diop*
- *the two persons claiming ownership of the land on which the unit will be located*

The technique used was interview surveys which ensured an in-depth and detailed exploration of the questions asked. These interviews were conducted on the basis of a semi-directive interview guide. In fact, this tool, through the intermediary of theme-questions, sets out the thread of the discussion between the researcher and the actors and acts as an intermediary for the exchange.

The local stakeholder meeting was conducted prior to the start of validation activities with VVB.

The comments by local stakeholders are invited in an open and transparent manner. A summary of the comments received has provided to the validation team along with the report indicating how due account was taken to the comments received.

The local stakeholder consultation process is mentioned in detail under section 7 of the Environmental and social impact study of the project /31/ which is checked by the validation team and found OK.

Validation team has checked the supportive document /31/ to confirm the stakeholder consultation and confirm that people were supportive to the project activity and expressed no potential negative comments on the project activity. As part of the validation, 4KES confirmed through document review that the stakeholders were invited in a transparent manner.

Since the comments of the stakeholders did not question any detail of the design of the project activity, no updates have been made with regard to the design of the project.

Further Validation team has checked the on-going communication with local stakeholders. By communicating all contact details of the project proponent and ESIA consultant during the meetings, stakeholders were informed about the creation of the grievance mechanism through, which any complaints, suggestions, recommendations etc. could be communicated. It mainly consists of complaint forms, which are available and can be filled at the village council of Taiba Ndiaye, as well as the sub-prefecture of Meouane as well as the regional government administration of Thiès. A formalized and official PETN Grievance Mechanism Procedure (GMP) is established as mentioned under section 2.2 Joint PD & MR /02/. There are other modes available for the comments as explained under section 2.2 Joint PD & MR /02/. PP has a well-established Grievance mechanism which was also verified during the PP and stakeholder interview. From the PP and stakeholder interview, Validation team has found that no negative comments have been received.

3.3.3 Environmental Impact

The project was subject to a Comprehensive Impact Assessment (EIA) which is mandatory for "projects that have limited environmental impacts and which can be mitigated by implementing measures or changes in design" in accordance with Decree Law 2001-01 of 15 January 2001 on the Environment Code (Article R40). Validation team has checked the Environmental and social impact study of the project activity /31/ which consist mainly of linking the sources of impacts associated with the project and the environmental components of the various receivers likely to be affected, with a view to identifying the negative impacts and proposing specific environmental measures to reduce them.

From the evaluation of the environmental and social impacts of the establishment of a park of 46 wind turbines in the municipality of Taïba Ndiaye, it appears clearly the positive arguments of such an investment militate in favour of the acceptability of this project. Indeed, in its quality of renewable energy which does not require any fuel, does not create greenhouse gases, does not produce toxic or radioactive waste, the introduction of the wind energy constitutes a credible alternative of diversification and reinforcement of the park of production of electricity in Senegal. Moreover, by fighting against the climate change, the wind energy takes part in the long term in the maintenance of the biodiversity of the natural environments.

In short, the above concludes in favor of the social and environmental acceptance of the project for the establishment of a wind farm of 46 turbines in the Commune of Taiba Ndiaye, in the department of Tivaouane.

Therefore, the project has been granted an Environmental Impact Assessment approval by decree on 21-June-2016 as verified from the supportive document /30/. Validation team has interviewed the local stakeholder and confirmed that project has positive impact on the environment and proper compensation was given to people.

3.3.4 Public Comments

In accordance with the requirement in clause 3.16.5 of the VCS standard version 4.1 “All VCS projects are subject to a 30-day public comment period. The date on which the project is listed on the project pipeline marks the beginning of the project’s 30-day public comment period”.

The PP listed their project activity in the VCS pipeline for 30 days from 20/10/2021 to 19/11/2021 (<https://registry.verra.org/app/projectDetail/VCS/2588>) for public comments.

No comments received during the commenting period, as evident from the VCS pipeline in the web interface.

3.3.5 AFOLU-Specific Safeguards

Not Applicable.

3.4 Application of Methodology

3.4.1 Title and Reference

The validation and verification team confirm that selected baseline and monitoring methodology is approved methodology and PP has correctly used it. All the applied methodology and tools are still valid.

- ACM0002 version 20.0 - “Grid-connected electricity generation from renewable sources”

Other guidelines and tools referenced in this methodology are:

TOOL23: “Additionality of first-of-its-kind project activities” version 03.0

TOOL07: “Tool to calculate the emission factor for an electricity system” version 7.0

ASB0034-2021: “Grid emission factor for West African Power Pool” version 01.0;

3.4.2 Applicability

Applicable criteria for the selected baseline methodology ACM0002, version 20.0 is assessed by the validation and verification team by means of document review & remote audit. The project applicability was confirmed against each condition in the approved methodology by the following methods.

- ACM0002 version 20.0 - “Grid-connected electricity generation from renewable sources”

ACM0002 version 20 applicability conditions	Project activity applicability	Validation Remarks
1. This methodology is applicable to grid-connected renewable energy power generation project activities that: a) Install a Greenfield power plant; b) Involve a capacity addition to (an) existing plant(s);	The project activity is a greenfield wind power plant substituting electricity produced on the grid	The project activity consists in the installation of a new grid connected renewable power plants at a site where

c) Involve a retrofit of (an) existing operating plant(s)/unit(s); d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s).		no renewable power plant was operated prior to the implementation of the project activity (Greenfield power plant). Hence point (a) is applicable. Completion certificates /17/, Production license /19/, Monthly electricity reports /21/, Photos/ video of site /25/, Signed PPA /26/ etc. have been checked to confirm the same. Project is supplying electricity to SENELEC. PPA /26/ has already signed between PP and SENELEC which has been verified by the validation team. Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.
2. The methodology is applicable under the following conditions: a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;	The project activity is the construction and operation of a wind power plant and hence the methodology is applicable	a) The project activity is the installation of a new wind power plant as verified during the remote interview. Completion certificates /17/, Technical specification /18/, Production license /19/, Photos/ video of site /25/, Signed PPA /26/ etc. have been checked to confirm the same.

		Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.
b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project activity does not involve any capacity additions, retrofits, rehabilitations or replacements	b) The project activity consists in the installation of a new grid connected renewable power plants at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield power plant). Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.
3. In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in an existing single or multiple reservoir, with no change in the volume of any of the reservoir or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7)³, is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the	Not applicable as the proposed project activity involves a wind power plant	The project activity is the installation of a new wind power plant. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.

³ Of applied methodology.

power density, calculated using equation (7), is greater than 4 W/m² d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: i. The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; ii. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; iii. Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 percent of the installed capacity of integrated hydro power project.		
4. In the case of integrated hydro power projects, project proponent shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration	Not applicable as the proposed project activity involves a wind power plant	The project activity is not an integrated hydropower plant. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.

has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity		
5. This methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site b) Biomass fired power plants/ units	The proposed project activity neither involves - switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site, nor - biomass fired power plants/units	It is a renewable energy project with no fuel-switch and biomass fired power plant involved. The project activity is the installation of a new wind power plant. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.
6. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project activity does not involve capacity additions, retrofits, rehabilitations or replacements	The project activity is the installation of a new wind power plant. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology ACM 0002 Version 20.0.
In addition, the applicability conditions included in the tools referred to above apply.	Applicability conditions of the applied tools are justified	The applicability conditions have been checked by the validation and the project was found in compliance.

- “Additionality of first of its kind project activities” version 03.0

No	Applicability condition	Applicability to this project activity	Validation Remarks
1	This methodological tool is applicable to project activities that wish to use the "first-of-its kind" approach to demonstrate additionality and that use versions of baseline and monitoring methodologies, or the "Tool for the demonstration and assessment of additionality" or the "Combined tool to identify the baseline scenario and demonstrate additionality", which allow using the "first-of-its-kind" approach for demonstrating additionality.	In accordance with ACM0002 methodology, additionality of the project activity is demonstrated and assessed using the latest version of the “TOOL23: Additionality of first-of-its-kind project activities” version 03.0	Justification provided by the PP is found OK. Detailed justification has been provided under section 3.4.5 of this validation report. Hence the project is in compliance with this tool.

- “Tool to calculate the emission factor for an electricity system” version 07.0

No	Applicability condition	Applicability to this project activity	Validation Remarks
1	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	As part of ACM0002, “operating margin” (OM), “build margin” (BM) and “combined margin” (CM) need to be estimated to calculate baseline emissions of the project activity that substitutes electricity in the Senegalese grid. Hence the tool is applicable.	Justification provided by the PP is found OK. Project is supplying electricity to SENELEC. PPA /26/ has already signed between PP and SENELEC which has been verified by the validation team. Hence the project is in compliance with this tool.
2	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off - grid power plants. In the latter case, two sub-options under	The emission factor for the project electricity system is calculated for grid power plants and off-grid power plants.	Justification provided by the PP is found OK.

No	Applicability condition	Applicability to this project activity	Validation Remarks
	the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 2: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Option IIb is applied, i.e. the tool is applicable.	Hence the project is in compliance with this tool.
3	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Since the project electricity system is not located partially or totally in an Annex I country - it is located in the Republic of Senegal - the tool is applicable.	Justification provided by the PP is found OK. Hence the project is in compliance with this tool.
4	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	There are no biofuels used in the project activity, i.e. the tool is applicable.	Justification provided by the PP is found OK. Hence the project is in compliance with this tool.

- Standardized Baseline “ASB0034-2021: “Grid emission factor for West African Power Pool” version 01.0”

No	Applicability condition	Applicability to this project activity	Validation Remarks
(a)	The project activity is implemented in any of the countries members of WAPP and is connected to the project electricity system;	The project activity is implemented in Senegal, member of WAPP, and connected to the same electricity system.	Justification provided by the PP is found OK. The project activity is implemented in Senegal, member of WAPP and connected to the project electricity system. Project is supplying electricity to SENELEC. PPA /26/ has already signed between PP and SENELEC which has been verified by the validation team. Hence the project is in compliance with this tool.
(b)	The CDM approved methodology that is applied to the project activity requires the determination of CO2 emission factor(s) through the application of the grid tool;	As part of ACM0002, “operating margin” (OM), “build margin” (BM) and “combined margin” (CM) need to be estimated to calculate baseline emissions of the project activity as per grid tool.	Justification has been provided by the PP and found OK. Hence the project is in compliance with this tool.
(c)	The project activity uses ex ante option for the grid emission factor as indicated in the tool i.e. no monitoring and recalculation of the emissions factor during the crediting period is required	Ex-ante option for grid emission factor is applied in this project activity.	Justification has been provided by the PP and found OK. Hence the project is in compliance with this tool.

VVB confirm that, the project applicability was checked against each condition in the approved methodology and applied tools and found confirming.

3.4.3 Project Boundary

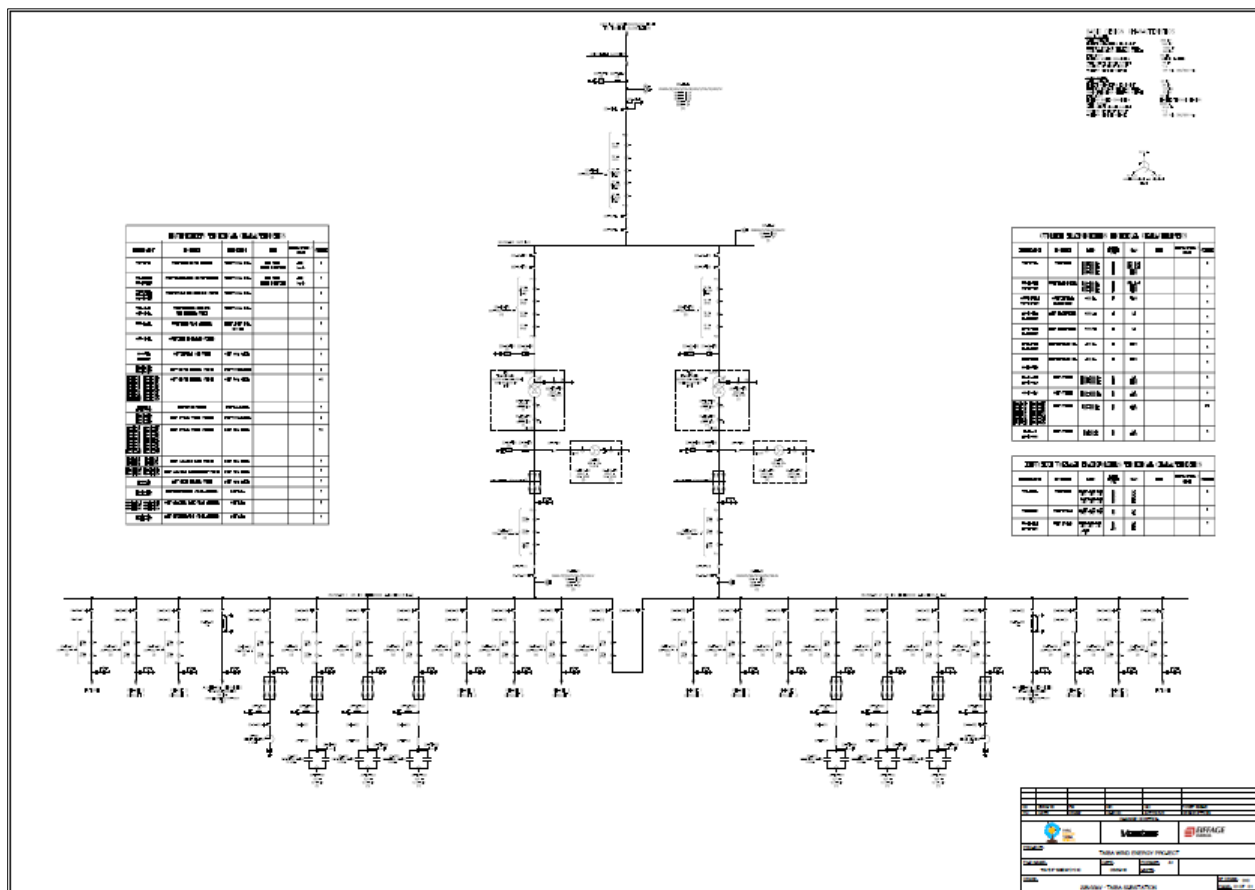
Para 20 of applied Large scale baseline methodology ACM0002 version 20.0 delineates the project boundary. As per the applied methodology, the project boundary is defined as.

“The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the CDM project power plant is connected to.”

The project boundary is therefore determined as:

- the project activity site, where the electricity is being produced,
- the grid that the power plant is connected to.

The project boundary diagram is as follows.



The following emission sources have been considered under the project boundary which is found OK according to the remote audit and the supportive documents.

Source		Gas	Included?	Justification/Explanation
Baseline	CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
		Other	No	Minor emission source
Project	For geothermal power plants, (fugitive) emissions of CH4 and CO2 from non-condensable gases contained in geothermal steam	CO ₂	No	Not applicable (Only for geothermal)
		CH ₄	No	Not applicable (Only for geothermal)
		N ₂ O	No	Not applicable
		Other	No	Not applicable
	For binary geothermal power plants, fugitive emissions of hydrocarbons such as n-butane and isopentane (working fluid) contained in the heat exchangers	Low GWP hydro-carbon/ refrigerant	No	Not applicable (Only for geothermal)
	CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO ₂	No	Not applicable (Only for solar thermal power plants and geothermal power plants)
		CH ₄	No	Not applicable
		N ₂ O	No	Not applicable
		Other	No	Not applicable
	For hydro power plants, emissions of CH4 from the reservoir.	CO ₂	No	Not applicable (Only for hydro)
		CH ₄	No	Not applicable
		N ₂ O	No	Not applicable
		Other	No	Not applicable

Opinion:

The validation team considers that GHG emissions as a result of the implementation of the proposed project activity which is not addressed by the applied methodology, i.e. ACM0002 version 20.0 is deemed to contribute less than 1% of the overall expected average annual emissions reductions. Validation team confirm that project boundary and selected sources, sinks and reservoirs are justified for the project.

3.4.4 Baseline Scenario

The project activity is the installation of a new grid-connected renewable power plant (Greenfield) as verified from the Completion certificates /17/, Production license /19/, Monthly electricity reports /21/, Photos/ video of site /25/, Signed PPA /26/ etc.. Hence according to para. 22 of applied methodology ACM0002 Version 20.0, the baseline scenarios is as follows:

“Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system.”

PP has selected the combined margin grid emission factor based on Standardized Baseline “ASB0034-2021: Grid emission factor for West African Power Pool (version 01.0)” (https://cdm.unfccc.int/methodologies/standard_base/2015/sb166.html) which is found OK. As per the standardized baseline /11/, the applicable grid emission factor value to calculate the emission reductions of the wind power plant project is 0.573 tCO₂/MWh which is found correct.

PP has correctly identified the baseline as per the applied methodology and emission reduction has been calculated accordingly.

Opinion:

The assessment team confirms:

- (a) The identified baseline scenario is justified.
- (b) The approved baseline methodology has been correctly applied to identify the most plausible baseline scenario and the identified baseline scenario reasonably represents what would occur in the absence of the proposed project activity.

3.4.5 Additionality

As per para 29 of the applied methodology ACM 0002, Version 20.0 /06/, The additionality of the project activity shall be demonstrated and assessed using the latest version of the “TOOL01: Tool for the demonstration and assessment of additionality”

PP has applied step 0 i.e. “Demonstration whether the proposed project activity is the first-of its-kind” of latest version of the tool “Methodological tool: Tool for the demonstration and assessment of additionality” Version 07.0.0 and additionality of the project activity is demonstrated and assessed using the latest version of the “TOOL23: Additionality of first-of-its-kind project activities” version 03.0 /13/.

As per applied “TOOL23: Additionality of first-of-its-kind project activities” version 03.0 /13/, Project activity is the first of its kind in the applicable geographical area if:

- (a) The project is the first in the applicable geographical area that applies a technology that is different from technologies that are implemented by any other project, which are able to deliver the same output and have started commercial operation in the applicable geographical area before the project design document is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier:

PP response: indeed, no other wind power has started operation in Senegal to date⁴.

Validation team response: Validation team has checked the Regulatory Commission for Electricity Sector website (<https://www.crse.sn/producteurs-independants>) and confirm that no other wind power has started operation in Senegal till date. Hence project qualify this para.

- (b) The project implements one or more of the measures;

PP response: indeed, the project implements wind power as a renewable energy measure.

Validation team response: Project activity is power generation based on renewable energy (wind energy) and hence implements one of the measures as per the tool TOOL23: Additionality of first-of-its-kind project activities” version 03.0 /13/. Hence project qualify this para.

- (c) The project participants selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal”.

PP response: indeed, the project participants selected a 10 years crediting period with no option of renewal.

Validation team response: Justification provided by the PP is found OK. Hence project qualify this para.

Hence Validation team confirm that this project activity is first-of-its-kind project activity and is thus additional.

3.4.6 Quantification of GHG Emission Reductions and Removals

The emission reduction (ERy) by the project activity during the crediting period is the difference between baseline emissions (BEy), project emissions (PEy) and emissions due to leakage (Ly). The calculations and formulae as addressed in the approved baseline and monitoring methodology ACM0002 Version 20.0 has been applied. All aspects related to the direct and indirect GHG emissions as relevant to the project activity have been addressed and are presented in a transparent manner, in line with the approved methodology.

⁴ <https://www.crse.sn/producteurs-independants>

Baseline emissions (BE_y)

Baseline emissions include only CO₂ emissions from electricity generation from fossil fuel fired power plants that are displaced due to the project activity, calculated as follows (as per para 39 of the applied methodology):

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE _y	Baseline emissions in year y (tCO ₂ /yr).
EG _{PJ,y}	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
EF _{grid,CM,y}	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”.

Calculation of EG_{PJ}

Because the project activity is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, then:

$$EG_{PJ,y} = EG_{facility,y} \text{ ((as per para 41 of the applied methodology))}$$

Where:

EG _{PJ,y}	= Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
EG _{facility,y}	= Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Therefore, the baseline emissions are calculated as follows:

$$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$$

Calculation of EF_{grid,CM,y}

Based on Standardized Baseline “ASB0034-2021: Grid emission factor for West African Power Pool” v1.0, the applicable grid emission factor value to calculate the emission reductions of the wind power plant project is 0.573 tCO₂/MWh. Validation team has checked the standardized baseline and found applicable for the project activity. Further Tool to calculate the emission factor for an electricity system (version 07.0) is used for this emission factor calculation which is in

compliance with the applied methodology i.e. ACM0002 Version 20.0 /06/. Hence the selected emission factor for the project activity is accepted to the validation team.

Project emission (PE_y)

According to para 31 of ACM0002, Version 20.0, the project emissions are calculated using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y	Project emissions in year y (tCO ₂ e/yr)
$PE_{FF,y}$	Project emissions from fossil fuel consumption in year y (tCO ₂ /yr)
$PE_{GP,y}$	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO ₂ e/yr)
$PE_{HP,y}$	Project emissions from water reservoirs of hydro power plants in year y (tCO ₂ e/yr)

The proposed project is a wind power plant that neither uses fossil fuel nor operates geothermal power plants or having water reservoirs (i.e. $PE_{FF,y} = 0$; $PE_{GP,y} = 0$; $PE_{HP,y} = 0$); therefore, the project emission is zero:

$$PE_y = 0$$

Leakage (L_y)

According to para 53 of ACM0002, Version 20.0, no leakage emissions are considered. The main emissions potentially giving rise to leakage in the context of electric sector projects are emissions arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport). These emissions sources are neglected.

Emission reductions (ER_y)

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y	Emission reductions in year y (t CO ₂ e/yr).
BE_y	Baseline emissions in year y (t CO ₂ e/yr)

PE_y Project emissions in year y (t CO₂e/yr).

Validation team confirm that the methodological choices/approaches and found OK. Validation team has verified the ex-ante ER calculation /04/ and found OK. The ex-ante Emission reduction is summarized below.

Ex-ante calculation of emission reductions:

	Value/Result	Source/reference
Total installed capacity	158.7 MW	Validation team has verified the value from production license /19/
Net electricity delivered to the grid (EG _{PJ,y})	449.8 GWh/yr	The value is verified DNV-Taiba Energy assessment report /34/
Grid emission factor (EF _{grid,CM,y})	0.573 tCO ₂ e/MWh	ASB0034 /11/
Baseline emissions (BE _y)	257,735 tCO ₂ e	Calculated
Project emissions (PE _y)	0 tCO ₂ e	Justified above.
Emission reductions (ER _y)	257,735 tCO ₂ e	ER _y = BE _y - PE _y

Summary of emission reductions

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
09/12/2019 – 08/12/2020	117,332	-	-	117,332
09/12/2020 – 08/12/2021	257,735	-	-	257,735

09/12/2021 – 08/12/2022	257,735	-	-	257,735
09/12/2022 – 08/12/2023	257,735	-	-	257,735
09/12/2023 – 08/12/2024	257,735	-	-	257,735
09/12/2024 – 08/12/2025	257,735	-	-	257,735
09/12/2025 – 08/12/2026	257,735	-	-	257,735
09/12/2026 – 08/12/2027	257,735	-	-	257,735
09/12/2027 – 08/12/2028	257,735	-	-	257,735
09/12/2028 – 08/12/2029	257,735	-	-	257,735
Total	2,436,950	-	-	2,436,950

Opinion:

As discussed above, the validation team confirms the following:

- All assumptions and data used by the project participants are listed in the PD, including their references and sources;
- All documentation used by project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PD;
- All values used in the PD are considered reasonable in the context of the proposed project activity;
- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, leakage and emission reductions;

All estimates of the baseline/project emissions can be replicated using the data and parameter values provided in the PD and corresponding spreadsheets.

Applied methodology and any referenced tools have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals during the project crediting period.

3.4.7 Methodology Deviations

No methodology deviations identified.

3.4.8 Monitoring Plan

The project activity uses the approved baseline and monitoring methodology ACM0002 (Version 20.0). Validation team considers the monitoring plan to be complying with the requirements of the methodology based on the following assessment. Details of the data collection and frequency of data recording and associated formats are described and observed to be adequate. The monitoring plan will give opportunity for real measurements of emission reductions that will be accrued.

The responsibilities and authorities for project management, procedures for monitoring and reporting, and QA/QC procedures have been systematically established and formalized. Data will be archived electronically until 2 years after the end of the crediting period.

Ex-ante parameter:

Following ex-ante parameter is reported under section 5.1 of VCS PD.

Data / Parameter	EF _{grid,CM,y}	VVB assessment
Data unit	tCO ₂ /MWh	The ex-ante parameter is as per the applied methodology ACM0002 (Version 20.0). Grid emission factor has been chosen from the standardized baseline /11/ which is a valid and hence accepted to the validation team. Based on Standardized Baseline “ASB0034-2021: Grid emission factor for West African Power Pool” v1.0 /11/, the applicable grid emission factor value to calculate the emission reductions of the wind power plant project is 0.573 tCO₂/MWh. Validation team has checked the standardized baseline and found applicable for the project activity. Further Tool to calculate the emission factor for
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”	
Source of data	As per ASB0034-2021	
Value applied:	0.573	
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” version 07.0	
Purpose of Data	Calculation of baseline emissions	

Comments	-	an electricity system (version 07.0) is used for this emission factor calculation which is in compliance with the applied methodology i.e. ACM0002 Version 20.0 /06/. Hence the selected emission factor for the project activity is accepted to the validation team.
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Ex-post parameter:

Following ex-post parameter is reported under section 5.2 of VCS PD /02/.

		WVB Assessment
Data / Parameter	EG_{facility,y}	The monitoring parameter, Unit, and descriptions are found OK.
Data unit	MWh/yr	
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	
Source of data	Measured directly with electricity meter(s) at substation.	Metering equipment is composed of two bi-directional meters for import/export (Model ITRON SL7000 class 0.2, main and back-up) located at SENELEC's Tobene sub-station delivery point as verified during the remote audit and photos /25/ which is in compliance with the applied methodology.
Description of measurement methods and procedures applied	Electricity outputs will be electronically stored and invoiced monthly to Senelec by PETN, after reconciliation with internal meter which readings are recorded on a record sheet by the Technical/Engineering/Maintenance Department under the Plant Manager's authority	The monitoring procedure is verified during the remote audit and found OK.

Frequency of monitoring/recording	Continuous measurement and at least monthly recording	The electricity export/import is continuous measured and monthly recorded which is in compliance with the applied methodology ACM0002 Version 20.0 /06/.
Value applied:	449,800	The estimated value is found OK. The value is verified DNV-Taiba Energy assessment report /34/.
Monitoring equipment	Bi-directional electronic electricity meters measuring the net electrical energy delivered to Senelec with the following specifications: Make: Itron Model: SL7000 Accuracy class: 0.2S ($\pm 0.2\%$) Serial ID: 1904406426-1904406427	Metering equipment is composed of two bi-directional meters for import/export (Model ITRON SL7000 class 0.2, main and back-up) located at SENELEC's Tobene sub-station delivery point as verified during the remote audit and photos /25/ which is in compliance with the applied methodology ACM0002 Version 20.0 /06/.
QA/QC procedures applied	Cross check of measurement results with invoiced electricity. Calibration of the meters will be carried out after each deviation of more than $\pm 0.5\%$ between the 2 meters as per manufacturer & PPA specifications.	Cross check of measurement results with invoiced electricity in accordance with the applied methodology ACM0002 Version 20.0 /06/. Calibration of the meters will be carried out after each deviation of more than $\pm 0.5\%$ between the 2 meters as per PPA specifications /26/.
Purpose of data	Calculation of baseline emissions	Validation team has checked and found OK
Calculation method	Electronic recording	Validation team has checked and found OK
Comments	EnergiExpress inspection/testing undertaken on 02/10/2020 as per report, valid until 02/10/2021	Validation team has checked and found OK

The monitoring plan and the QA/QC procedure would ensure the accurate monitoring of the data/parameters mentioned in the VCS PD throughout the implementation of the project activity. The validation team confirms that the monitoring procedures mentioned are best as per Industry norms.

Opinion: The assessment team confirms that:

- (a) The monitoring plan mentioned in VCS PD is in compliance with the applied methodology ACM0002 Version 20.0 and any referenced tools;
- (b) Based on the remote audit, the assessment team confirms that monitoring arrangements described in the monitoring plan are feasible within the project design;
- (c) Based on the remote interview with the PP, assessment team confirms that the project participant deputed the competent personal to execute the monitoring approach and to follow the monitoring plan.

3.5 Non-Permanence Risk Analysis

The project activity does not require a non-permanence risk analysis. Hence this is not applicable.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

VVB was able to confirm that the monitoring plan contained in Joined PD & MR /02/ is in accordance with the approved large scale methodology applied for the project activity i.e. “ACM0002 ver. 20 /06/ and its applicable tools.

The parameter stated in the monitoring plan /02/ and the applied methodology /06/ has been fulfilled in the current monitoring period.

The GHG emission reductions were correctly calculated on the basis of the applied methodology ACM0002 Version 20.0 /06/ and the formulae given in the Joined PD&MR /02/.

Parameter monitored:

Verification team confirms through remote audit verification and from the document review, the actual monitoring system complies with the monitoring plan mentioned in the Joined PD&MR /02/. According to the monitoring plan in the Joined PD&MR /02/, there is 1 monitoring parameter required to be monitored.

4.1.1 $EG_{\text{facility},y}$ (MWh)

	VVB Assessment
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Data / Parameter	EG _{facility,y}				The monitoring parameter, Unit and descriptions are as per the Joint PD&MR /02/.								
Data unit	MWh/yr												
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y												
Value applied:	<table><tr><td>2019</td><td>2020</td><td>2021</td><td>Total</td></tr><tr><td>22,936</td><td>216,520</td><td>297,960</td><td>537,416</td></tr></table>				2019	2020	2021	Total	22,936	216,520	297,960	537,416	The electricity export/import is continuous measured and monthly recorded by energy meters. Metering equipment is composed of two bi-directional meters for import/export (Model ITRON SL7000 class 0.2, main and back-up) located at SENELEC's Tobene sub-station delivery point as verified during the remote audit, Single line diagram /20/ and photos /25/. Electricity output is electronically stored and invoiced monthly to Senelec by PETN, after reconciliation with internal meter which readings are recorded on a record sheet by the Technical/Engineering/ Maintenance Department under the Plant Manager's authority. The electricity supplied to grid is continuously measured and monthly recorded in the form of the monthly electricity reports /21/ as verified during the remote audit. The frequency of monitoring/recording is in accordance with the Joint PD&MR /02/. Verification team has verified the reported electricity data with the monthly electricity reports /21/ and found consistent. The monthly electricity data is also cross-checked from the monthly invoices /22/ and found consistent. The purpose of the monitoring parameter is to calculate the baseline emission which is in accordance with the Joint PD&MR /02/.
	2019	2020	2021	Total									
	22,936	216,520	297,960	537,416									
Comments	✓ Main & check meters SL7000 ITRON #: 1904406426-1904406427 class 0.2S at delivery point ✓ Meters readings difference confirmed as <0.5% every month												

	✓ Initial meters test report by Itron has been provided (dated 01/08/2019, but commissioned on 09/12/2019) while EnergiExpress inspection/testing was undertaken on 02/10/2020 as per report, valid until 02/10/2021	Verification team has checked the meter details during the remote audit, photos /25/ and calibration reports /23/ and found consistent with the details mentioned in the Joint PD&MR /02/. During the monitoring period, no replacement of meter was observed. As per the PPA /26/, Calibration of the meters will be carried out after each deviation of more than +- 0.5% between the 2 meters. VVB confirm that meters readings difference was <0.5% every month for this monitoring period, hence no further calibration needed. Initial calibration of the energy meters /23/ has been checked by the VVB and found OK.
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Opinion: The verification team confirms;

- The monitoring plan has been implemented as per the Joint PD&MR /02/;
- The monitoring complies with the requirement of the applied methodology /06/;
- The information inflow (from data generation, aggregation, to recording, calculation and reporting) is included above under each parameters and confirms to the requirement of the Joint PD&MR /02/;
- The values included in the monitoring report and corresponding emission reduction sheets are verified and included under monitoring parameter, wherever appropriate;

Parameters not monitored/ex-ante:

4.1.2 EF_{grid,CM,y} (tCO₂/MWh)

	Discussion and verification assessment
Purpose of data	Calculation of baseline emissions
Verified value	0.573 tCO ₂ /MWh
Source of value	Grid emission factor has been chosen from the standardized baseline /11/.
Justification	It is fixed for the entire crediting period as per the Joint PD&MR /02/. The value of the combined margin emission factor has been correctly taken as per the Joint PD&MR /02/ and Hence accepted by the verification team.

Opinion:

In the opinion of assessment team, emission factor value that was applied in the calculations is consistent with Joint PD&MR /02/ and correctly emission reduction spread sheet /04/ and justified.

Assessment of Data and Calculation of GHG emission reductions:

Verification team has checked the emission reduction calculation /04/ for this monitoring period 09/12/2019 to 31/07/2021 (including both the days) and found correct and in compliance with the Joint PD&MR /02/ and applied methodology ACM0002 Version 20.0 /06/. Summary of the baseline emissions, project emissions, leakage emissions and net emission reductions are as follows:

Baseline Emissions:

Year	BE _y	EF _{grid,CM,y}	EG _{facility,y}
	t CO ₂	tCO ₂ /MWh	MWh
2019 (from 09/12/2019)	13,142	0.573	22,936
2020	124,066	0.573	216,520
2021 (until 31/07/2021)	170,731	0.573	297,960
TOTAL	307,939		537,416

Project Emissions:

No project emissions are considered as the project activity only involves renewable electricity generation from the wind power plant. Hence according to ACM0002 guidelines, PE_y = 0.

Leakage emissions:

As stated in the applicable methodology, no leakage emissions are considered, therefore LE_y = 0.

Net GHG Emission reductions for the monitoring period 09/12/2019 to 31/07/2021 (including both the days):

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2019	13,142	-	-	13,142
2020	124,066	-	-	124,066
2021	170,731	-	-	170,731
Total	307,939	-	-	307,939

Opinion:

The verification team confirms that

- The complete data set for the identified and required parameter for the operational days in the current monitoring period was available;
- The reported data has been cross checked with available records, as indicated in the section 4.1 above under monitored data, wherever appropriate;
- The baseline, project and leakage emissions have been determined in accordance with the requirement of the applied methodology, as contained in the final Joint PD&MR /02/ and corresponding emission reductions spreadsheet;
- The assumptions, emission factors and default values used are justified, as indicated in the section 4.1 above.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

Evidences (Documents/ Microsoft Teams interviews) referred for verification of monitoring parameter and fixed parameter are defined under section 4.1. We further confirm that, sufficient evidence covering the entire monitoring period and at the required frequency were available. A list of referred documents for verification is also included in Appendix 1 of this report.

5 VALIDATION AND VERIFICATION CONCLUSION

AERA Group has commissioned “4K Earth Science Private Limited” to carry out the Joint Validation and Verification of the project “Taiba N'Diaye Wind Power”, with regard to the relevant requirements of VCS Standard Version 4.1. The monitoring period verified is from 09/12/2019 to 31/07/2021 (including both the days).

The project activity involves the installation and operation of a 158.7 MW wind farm in the commune of Taiba N'Diaye, Senegal. The project activity which is Senegal's first utility-scale wind farm, consists of 46 Vestas V126 wind turbines, for a total installed capacity of 158.7 MW. They utilise a 117m tubular steel tower and a blade length of 61.7m giving a very large swept area of 12,469 m², allowing the wind turbines to maximise the amount of energy captured from the wind, and are designed for at least 20 years of operation.

The main elements of the wind power plant are Vestas V126-3.45 MW turbines, pitch regulated upwind turbines with active yaw and a three blade rotor. The wind turbine family utilises the OptiTip® concept and a power system based on an induction generator and full-scale converter. With these features, the wind turbine is able to operate the rotor at variable speed and thereby maintain the power output at or near rated power even in high wind speed. The technical details of the project is verified from the completion certificate /17/, technical specification of wind turbines /18/, single line diagram /20/, signed PPA /26/, photos /25/ and during remote audit and found consistent.

The project has been built by Lekela and is owned and operated by Parc Eolien Taiba N'Diaye S.A.U.

PETN is designed to generate electricity for at least 20 years through its 46 Vestas wind turbines. The generated electricity is supplied to SENELEC, the state-owned utility, under a long-term power purchase agreement (PPA) /26/.

In the absence of the project activity, electricity would have been generated by the operation of grid connected power plants, which are predominantly fossil fuel based. Therefore Taiba N'Diaye Wind Power project #2588 displaces carbon dioxide that would have been produced in the absence of the project activity by the grid connected power plants.

It has been estimated that approximately 450 GWh of electricity will be generated annually by the project (equivalent to 2,834 hours/year or 32% load factor), displacing approximately 243,695 tonnes of carbon dioxide (tCO₂) per annum and over 2.4 million tCO₂ over the 10-years crediting period.

Project is fully commissioned as verified from the completion certificates /17/.

Validation and Verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified emission reductions (VERs).

A risk-based approach has been followed to perform this joint validation and verification. In the course of the joint validation and verification, 03 Corrective Action Requests (CAR) and 04 Clarification Requests (CL) were raised and successfully closed out. 00 Forward action request (FARs) was raised.

The joint validation and verification is based on the Joint PD & MR, additional documents related to baseline and monitoring methodology, ER calculation sheet, the subsequent background investigation, follow-up interviews and supporting documents made available to the validation and verification team by project proponent.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. As a result of the validation, the validation team confirms that:

- The project fulfils criteria of VCS Standard Version 4.1.

- The project is in line with all relevant VCS requirements.
- The project additionality is sufficiently justified in the Joint PD & MR.
- The monitoring plan is transparent, adequate and in line with applied baseline and monitoring methodology ACM0002 version 20.0.
- The calculation of the Baseline emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 2,436,950 tCO₂e is most likely to be achieved within the 10 years of crediting period.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria.

No restrictions or uncertainties were identified related to the joint validation and verification.

The management of the 'Parc Eolien Taiba N'Diaye S.A.U' responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions on the basis set out within the project final JOINT PROJECT DESCRIPTION & MONITORING REPORT Version 1.0 dated 26/08/2021. The calculation and determination of GHG emission reductions from the project is the responsibility of the management of the Aera Group. The development and maintenance of records and reporting procedures are in accordance with the JOINT PROJECT DESCRIPTION & MONITORING REPORT Version 2.1 dated 14/01/2022.

It is our responsibility to express an independent GHG verification opinion on the GHG emissions and on the calculation of GHG emission reductions from the project for the period 09/12/2019 to 31/07/2021 based on the reported emission reductions in the final JOINT PROJECT DESCRIPTION & MONITORING REPORT Version 2.1 dated 14/01/2022 for the same period.

Further during this monitoring period (09-December-2019 to 31-July-2021), the project has produced a net total of 537,416 MWh electricity and a total amount of 307,939 tCO₂ of emission reduction. The monitoring period subject to this monitoring report is inclusive of first and last day of period.

Based on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these, 4KES planned and performed our work to obtain the information and explanations that we considered necessary to provide sufficient evidence for us to give reasonable assurance that this reported amount of GHG emission reductions for the period is fairly stated.

4KES confirms the following;

Verification period: From [09-December-2019] to [31-July-2021]

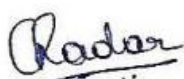
Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2019 (from 09/12/2019)	13,142	0	0	13,142
2020	124,066	0	0	124,066
2021 (until 31/07/2021)	170,731	0	0	170,731
Total	307,939	0	0	307,939

Validation and Verification team confirm that, project complies with the validation and verification criteria for projects and their GHG emission reductions or removals set out in VCS Version 4.

Therefore, 4KES is able to certify that the project is in full compliance with the VCS Standard Version 4.1 and recommend registration and issuance of the project activity.

Approved by



Chandrakala .R

Director

Date: 15/01/2022

4K Earth Science Pvt. Ltd

Place: Bangalore, India

APPENDIX 1: REFERENCES

S. No.	Name of document	date
/1/	Draft JOINT PROJECT DESCRIPTION & MONITORING REPORT	Version 1.0, dated 26/08/2021
/2/	Final JOINT PROJECT DESCRIPTION & MONITORING REPORT	Version 2.1, dated 14/01/2022
/3/	Draft ER spread-sheet	Corresponding to PD Version 1.0
/4/	Final ER spread-sheet	Corresponding to PD Version 2.1
/5/	CDM-EB: Clean Development Mechanism Validation and Verification Standard for project activity	Version 03.0
/6/	Approved CDM Methodology ACM0002: Grid-connected electricity generation from renewable sources - Version 20.0	-
/7/	VCS Verified Carbon Standard: VCS Program Guide, VCS Version 4.0	-
/8/	VCS Verified Carbon Standard: VCS Standard, v4.1	-
/9/	VCS Verified Carbon Standard: Registration and Issuance Process Version 4.0	-
/10/	VCS Verified Carbon Standard: Validation and Verification Manual, Version 3.2	-
/11/	ASB0034-2021: Grid emission factor for West African Power Pool (version 01.0) https://cdm.unfccc.int/methodologies/standard_base/2015/sb_166.html	From 24/3/2021
/12/	TOOL07: "Tool to calculate the emission factor for an electricity system" version 7.0	-
/13/	TOOL23: "Additionality of first-of-its-kind project activities" version 03.0	-

/14/	Certificate of Incorporation of “Parc Eolien Taiba N'Diaye S.A.U”	-
/15/	Declaration for double counting	01/09/2021
/16/	CDM project activity deregistration request form	04/11/2021
/17/	Completion certificate for the project between PP and SENELEC	Dated 09/12/2019 (Group 1), 06/11/2020 (Group 2) and 30/06/2021 (Group 3)
/18/	Technical specification of the Wind turbines and energy meters	-
/19/	Transmission decree granting license by Ministry of energy and energy development renewables	08/08/2017
/20/	Single line diagram showing metering location	-
/21/	Monthly electricity report between PP and SENELEC for the monitoring period 09/12/2019 to 31/07/2021	-
/22/	Monthly invoices or the monitoring period 09/12/2019 to 31/07/2021	-
/23/	Initial test certificate by Itron Global Trading dated 01/08/2019 Energi Express inspection/testing undertaken on 02/10/2020, valid until 02/10/2021	-
/24/	Trainings to employees and others (youth)	-
/25/	Photographic/video evidence of project site along with meter and transmission	-
/26/	Signed PPA between PP and SENELEC	31/12/2013, amendment 10/10/2016
/27/	1. Grievance mechanism dated SEPTEMBER 2016 2. Grievances and their closure	-
/28/	Evidence documents related to local stakeholder meeting • Announcement note • Announcement in local news paper • Screenshots of meeting • Attendance sheet • Summary of meeting	-
/29/	Supportive for the sustainable development	2020 annual report

	Employment	
/30/	Order approving the environmental impact study by the environment authority	21/06/2016
/31/	REVISED ENVIRONMENTAL AND SOCIAL IMPACT STUDY OF THE WIND FARM PROJECT IN TAIBA NDIAYE	04/02/2016
/32/	Organization structure with responsibility	-
/33/	Service and availability agreement between PP and VESTAS SENEGAL SARLU	Dated 23/02/2017
/34/	Taiba N'Diaye Wind Power: Energy Assessment Report (Report No.: 159858-ZACT-R-02, Issue B) by DNV	08/07/2016

APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

Table 1. CL from this validation

CL ID	01	Section no.	3.1	Date: 29/10/2021
Description of CL				
From the remote audit and the documents review, Validation team has found that installed capacity of individual wind turbine is 3.45 MW. Under section 1.1 of the "Joint Project Description & Monitoring Report", PP has mentioned "the first phase with the installation of 16 wind turbines i.e. a total power of 52.8 MW". PP need to clarify how 16 turbines comes to a capacity of 52.8 MW.				
Project participant response				Date: 02/11/2021
Phase-wise capacities have been corrected to 55.2 + 55.2 + 48.3 MW				
Documentation provided by project participant				
Updated Joint PD&MR				
DOE assessment				Date: 20/11/2021
Corrections have been done in the revised Joint PD&MR /02/ and found consistent with technical specification of the wind turbines /18/. Hence this CL is closed.				

CL ID	02	Section no.	3.1	Date: 29/10/2021
Description of CL				
Under section 1.7 of the “Joint Project Description & Monitoring Report”, PP has mentioned “Parc Eolien Taiba N’Diaye S.A.U has gained project ownership through a production license and power purchase agreement with Senegalese public utility SENELEC, while AERA Group has gained ownership through a carbon certification and trading agreement with SENELEC.”. PP need to clarify how AERA Group has gained ownership through a carbon certification and trading agreement with SENELEC as SENELEC is just power purchaser.				
Project participant response				Date: 02/11/2021
The complete chain of ownership is PETN > SENELEC > AERA, because PETN – SENELEC PPA involves a clause of environmental attributes cession to SENELEC (as per PPA §8.3.6 about ‘revenus issus des attributs environnementaux’ p. 26, which in turns entered into ERPA agreement with AERA.				
Documentation provided by project participant				
Updated Joint PD&MR C35 - 20131213 - CAE signé Taiba and Amendments 1 & 2 (PPA) ERPA				
DOE assessment				Date: 20/11/2021
Clarification has been provided by the PP and found OK. Validation team has checked the PPA /26/to confirm the same. Hence this CL is closed.				

CL ID	03	Section no.	3.4.6	Date: 29/10/2021
Description of CL				
Under section 4.4 of the “Joint Project Description & Monitoring Report”, PP has mentioned the Net electricity delivered to the grid as 451,195 MWh/yr however no source is defined. PP need to clarify the source of the data.				
Project participant response				Date: 03/11/2021
Expected ex-ante power generation is derived from DNV 20160708-DNV-Taiba Energy assessment report, actually 449.8 GWh/yr Evidence for total capacity has also ben referenced from Production licence.				
Documentation provided by project participant				
20160708-DNV-Taiba Energy assessment report 2017 08 08 - MEDER lett transmission Licence de production PETN				
DOE assessment				Date: 20/11/2021

Corrections have been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. The power generation is verified from Energy assessment report /34/ prepared by third party DNV and the project capacity has been verified from production license /19/ by Ministry of energy and energy development renewables. Hence this CL is closed.

CL ID	04	Section no.	3.4.8	Date: 29/10/2021
Description of CL				
Under the monitoring parameter “EG _{facility,y} ” (section 5.2 of the VCS PD), PP has mentioned that “Calibration of the meters will be carried out after each deviation of more than +/- 0.5% between the 2 meters as per manufacturer & PPA specifications.” However during the remote audit, it was found that calibration is done annually. PP need to clarify along with supportive.				
Project participant response				Date: 02/11/2021
As per PPA metering annex 12 extract §8.4.2 the meters indeed need to be inspected and tested annually, yet only re-calibrated if deviation of more than +/- 0.5%				
Initial meters test report by Itron has been provided (dated 01/08/2019, but commissioned on 09/12/2019) while EnergiExpress inspection/testing was undertaken on 02/10/2020 as per report, valid until 02/10/2021				
Documentation provided by project participant				
Inspection compteurs - extrait PPA annexe 12				
DOE assessment				Date: 20/11/2021
The response provided by the PP is found OK. PPA /26/ has been checked and found that calibration is required for deviation of more than +/- 0.5%. Hence this CL is closed.				

Table 2. CAR from this validation

CAR ID	01	Section no.	3.1	Date: 29/10/2021
Description of CAR				
1. The project title mentioned on page 1 of the “Joint Project Description & Monitoring Report” is not consistent within page 1.				
2. In the “Joint Project Description & Monitoring Report”, PP has mention solar project on no. of places which need to be corrected.				
Project participant response				Date: 02/11/2021
1. Taiba N'Diaye Wind Power is the Project title, as per following table Project description & monitoring report is the document's title, on a separate line.. ('Joint' added)				
2. Erroneous 'solar' references have been corrected to 'wind'				
Documentation provided by project participant				

Updated Joint PD&MR	
DOE assessment	Date: 20/11/2021
1. Correction has been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. Hence this part of CAR is closed. 2. Corrections have been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. Hence this part of CAR is closed.	

CAR ID	02	Section no.	3.1	Date: 29/10/2021
Description of CAR				
1. Under section 1.2 of the “Joint Project Description & Monitoring Report”, PP need to clearly mention whether the project is a grouped project (as per template requirements). 2. Under section 1.13 of the “Joint Project Description & Monitoring Report”, PP need to “Describe the conditions existing prior to project initiation and demonstrate that the project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction. Where the baseline scenario is the same as the conditions existing prior to the project initiation, there is no need to repeat the description of the scenarios (rather, just state that this is the case and refer the reader to Section 2.4 (Baseline Scenario).” (as per template requirements). 3. Under section 1.15.1 of the “Joint Project Description & Monitoring Report”, PP need to clearly mention Where the project has been registered under any other GHG program, provide the registration number and details (as per template requirements). 4. Under section 2.2 of the “Joint Project Description & Monitoring Report”, PP need to “Describe the process for, and the outcomes from, the local stakeholder consultation conducted prior to the joint validation and verification. Include details on the following: • The procedures or methods used for engaging local stakeholders (e.g., dates of announcements or meetings, periods during which input was sought). • The procedures or methods used for documenting the outcomes of the local stakeholder consultation. • The mechanism for on-going communication with local stakeholders. • How due account of all and any input received during the consultation has been taken. Include details on any updates to the project design or justify why updates are not appropriate” (as per template requirements). 5. Under section 2.4 of the “Joint Project Description & Monitoring Report”, PP need to “Demonstrate how due account of all and any comments received during the public comment period has been taken.				

Include details on any updates to the project design or demonstrate the insignificance or irrelevance of comments” (as per template requirements).	
Project participant response	Date: 02/11/2021
1. Section 1.2 updated (not a grouped project) 2. Section 1.13 updated (The baseline scenario is the same as the conditions existing prior to the project initiation; cf. Section 3.4 (Baseline Scenario).) 3. Section 1.15.1 updated (¹ As per CDM Project 5846 : Taiba N’diaye Wind Energy project, Senegal registered on 29 Feb 12, undergoing de-registration) 4. Correction done 5. Section 2.2 Public comments period completion	
Documentation provided by project participant	
Updated Joint PD&MR	
DOE assessment	Date: 20/11/2021
1. Correction has been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. Hence this part of CAR is closed. 2. Correction has been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. Hence this part of CAR is closed. 3. Correction has been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. Validation team has checked the UN webpage (https://cdm.unfccc.int/Projects/DB/DNV-CUK1330446091.13/view) and confirm that project is not eligible for registration. Further project CDM deregistration form /16/ is also submitted to UNFCCC. Hence this part of CAR is closed. 4. Correction has been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. Hence this part of CAR is closed. 5. Correction has been done in the revised “Joint Project Description & Monitoring Report” /02/ and found OK. No comment has been received during the public comment period. Hence this part of CAR is closed.	

CAR ID	03	Section no.	3.1	Date: 29/10/2021
Description of CAR				
During the remote interview, Validation team has found that project is fully commissioned however the same is not clearly reported under section 1.11 of the “Joint Project Description & Monitoring Report”. PP need to report the commissioning date of each phase.				
Project participant response				Date: 02/11/2021
Phase-wise commissioning dates included				
Documentation provided by project participant				

Updated Joint PD&MR	
DOE assessment	Date: 20/11/2021
Corrections have been done in the revised "Joint Project Description & Monitoring Report" /02/ and found OK. The commissioning dates have been checked from the completion certificate /17/ and found consistent. Hence this CAR is closed.	

Table 3. FAR from this validation

No FAR is raised during this validation

FAR ID	**	Section no.	Date: DD/MM/YYYY
Description of FAR			
Project participant response			Date: DD/MM/YYYY
Documentation provided by project participant			
DOE assessment			Date: DD/MM/YYYY

APPENDIX 3: COMPETENCE OF TEAM MEMBERS

Certificate of Competence						
Name	☒ Mr. ☐ Ms.	Chetan Swaroop Sharma				
Qualification Procedure	Fulfils the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GHG Projects.					
Appointed to work as:						
	CDM Validator/Verifier	Team Leader	Team Member	Technical Expert	Technical Reviewer	Financial Expert

Appointed	Yes	Yes	Yes	Yes	Yes	No
Appointed Date	27-04-2021					
Authorized to work as Technical Expert for:						
Authorized Technical Area	Sectoral Scope	TA Code	Technical Area within the scope			
	Energy industries (renewable - / non-renewable sources)	1.1	Thermal energy generation			
	Energy industries (renewable - / non-renewable sources)	1.2	Renewables			
	Energy distribution	2.1	Energy distribution			
	Energy demand	3.1	Energy demand			
	Waste handling and disposal	13.1	Solid waste and wastewater			
	Waste handling and disposal	13.2	Manure			
Authorized to work as Local Expert for:						
Country/Countries	India					
Compliance check by: Anand S. R.						

<u>Certificate of Competence</u>						
Name	☒ Mr. ☐ Ms.	Ma Paa Puratchikkanal				
Qualification Procedure	Fulfils the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GHG Projects.					
Appointed to work as:						
	CDM Validator/Verifier	Team Leader	Team Member	Technical Expert	Technical Reviewer	Financial Expert
Appointed	Yes	Yes	Yes	Yes	Yes	No

Appointed Date	27-04-2021		
Authorized to work as Technical Expert for:			
Authorized Technical Area	Sectoral Scope	TA Code	Technical Area within the scope
	Energy industries (renewable - / non-renewable sources)	1.1	Thermal energy generation
	Energy industries (renewable - / non-renewable sources)	1.2	Renewables
	Energy demand	3.1	Energy demand
	Construction	6.1	Construction
	Waste handling and disposal	13.1	Solid waste and wastewater
	Waste handling and disposal	13.2	Manure
	Agriculture	15.1	Agriculture
Authorized to work as Local Expert for:			
Country/Countries	India and Sri Lanka		
Compliance check by: Anand S. R.			

APPENDIX 4: ABBREVIATIONS

CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
EB	Executive Board
EISA	Environmental and Social Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request

GHG	Greenhouse Gases
IPCC	Intergovernmental Panel for Climate Change
ISO	International Organization for Standardization
MP	Monitoring Period
MR	Monitoring Report
MW	Mega Watt
MWh	MegaWatt hour
PD	Project Description
PP	Project proponent
QA/QC	Quality Assurance/Quality Control
tCO ₂	Tonnes of Carbon Dioxide
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit